LESSON 2.6

Divide Multi-Digit Numbers Using an Algorithm

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can divide multi-digit whole numbers and explain the meaning of the quotient and remainder.

WHY THIS MATTERS The space station has 1,344 nutrition bars that must be divided equally among 12 crew sections for a 4-week rotation. How many bars does each section receive?

REMEMBER

Divide Multi-Digit Numbers Using an Algorithm

Algorithm: A step-by-step set of rules or instructions to solve a math problem. Long Division Algorithm: D (Divide), M (Multiply), S (Subtract), B (Bring Down). Formula: Divisor) Dividend = Quotient R Remainder Dividing means splitting a total into equal groups. Long division finds the quotient one digit at a time by repeating the same four steps: divide, multiply, subtract, bring down.

1. D - Divide: write quotient digit above the bar.
2. M - Multiply: digit × divisor.
3. S - Subtract: find the difference.
4. B - Bring down: next digit from dividend.
5. Repeat: continue until no digits remain (leftover = remainder). Example: 1,344 ÷ 12 = 112

Words: Algorithm *(Algoritmo)* · Dividend *(Dividendo)* · Divisor *(Divisor)* · Quotient *(Cociente)* · Remainder *(Residuo)*

PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is 7 × 40?

(A) 28
(B) 47
(C) 280
(D) 2,800

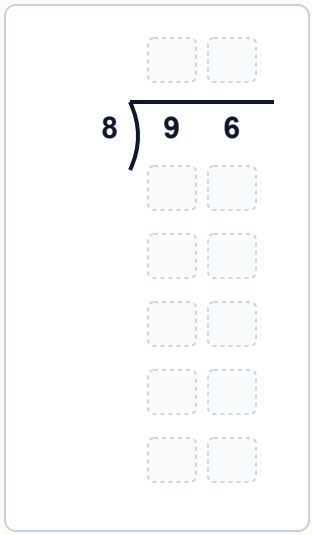
SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

2 **Circle the letter of the best answer. Show how you know.**

What is 96 ÷ 8?

(A) 8
(B) 11
(C) 12
(D) 14



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Which multiplication fact helps you divide 200 by 5?

(A) 5 × 40 = 200
(B) 5 × 20 = 100
(C) 5 × 4 = 20
(D) 5 × 50 = 250

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

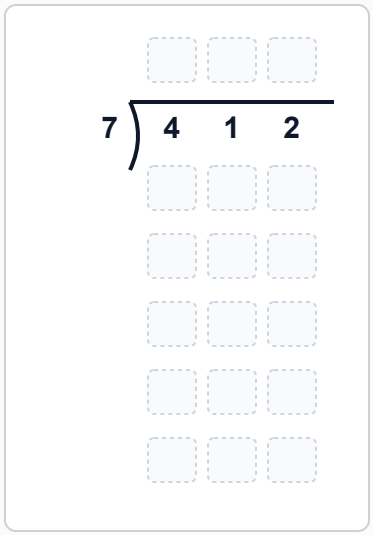
Estimate 412 ÷ 7 by using friendly numbers.

(A) 400 ÷ 8 = 50

(B) 400 ÷ 4 = 100

(C) 410 ÷ 10 = 41

(D) 420 ÷ 7 = 60



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

5 **Circle the letter of the best answer. Show how you know.**

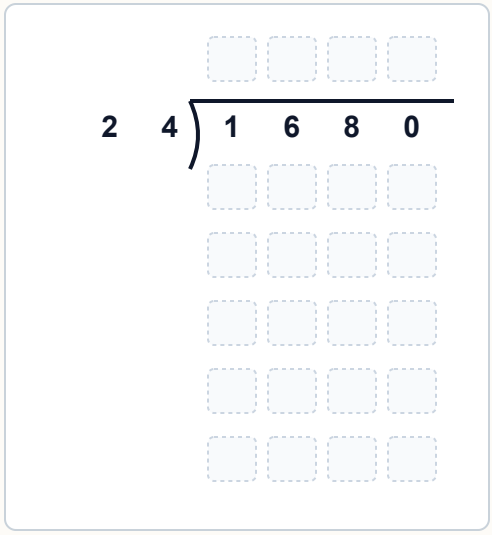
A school is ordering 1,680 pencils to distribute equally to 24 classrooms for the school year. 1,680 ÷ 24 = ?

(A) 60

(B) 70

(C) 72

(D) 80

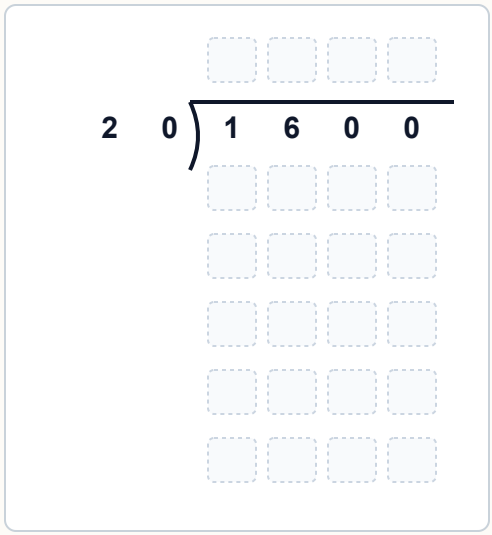


1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

6 **Circle the letter of the best answer. Show how you know.**

A school is ordering 1,680 pencils to distribute equally to 24 classrooms for the school year. Estimate first: rounding to 1,600 ÷ 20 gives about how many pencils per room?

(A) 40
(B) 60
(C) 80
(D) 100



1. **Divide** *How many fit?*
2. **Multiply** *Multiply back.*
3. **Subtract** *What is left?*
4. **Bring down** *Next digit.*

EXPLAIN YOUR THINKING

You ran the cycle three times. What is 2,184 ÷ 14, and how did you know when to stop repeating divide, multiply, subtract, bring down?

SENTENCE STARTER 2,184 ÷ 14 = ___ because I repeated divide, multiply, subtract, and bring down until there were no digits left to bring down, and what was left at the end was ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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show every step

1 **Circle the letter of the best answer. Show how you know.**

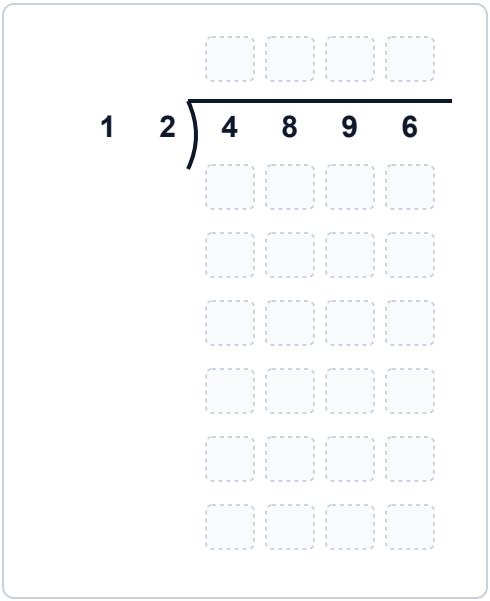
What is 4,896 ÷ 12?

(A) 48

(B) 380

(C) 408

(D) 480



2 **Circle the letter of the best answer. Show how you know.**

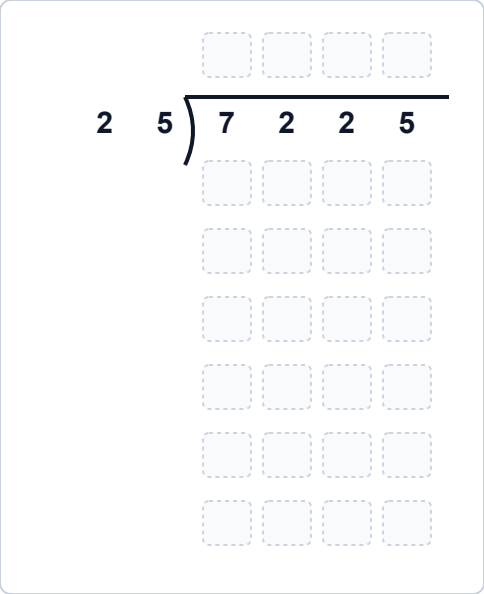
What is 7,225 ÷ 25?

(A) 29

(B) 289

(C) 389

(D) 2,890



3 **Circle the letter of the best answer. Show how you know.**

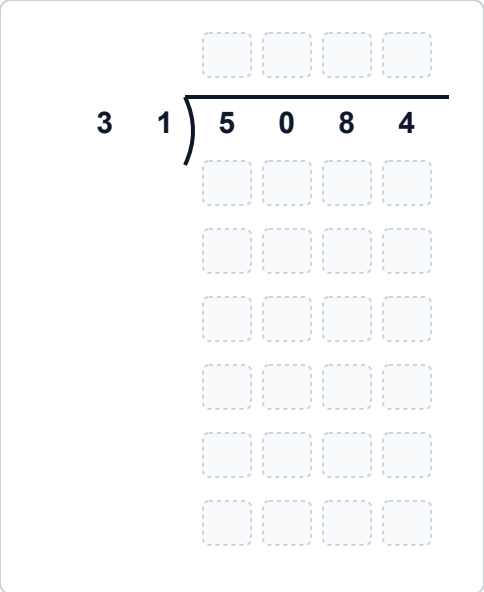
What is 5,084 ÷ 31?

(A) 154

(B) 164

(C) 164 r2

(D) 174



PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A school is ordering 1,680 pencils to distribute equally to 24 classrooms for the school year. If the pencils were split among 20 classrooms instead of 24, would each room get more or fewer?

(A) Fewer, because 20 is less than 24

(B) The same, because the total did not change

(C) More, because the pencils are split among fewer rooms

(D) You cannot tell

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A school orders 3,648 pencils packed 24 to a box. How many boxes are there?

(A) 144

(B) 148

(C) 152

(D) 160

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

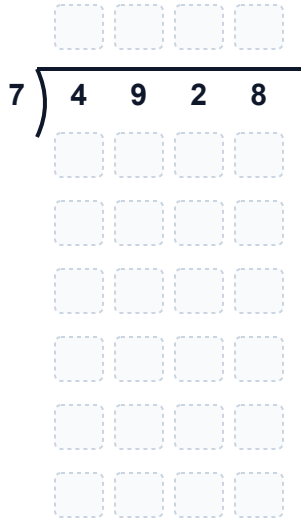
Path 1: The first arcade door shows 4,928 ÷ 7. Which quotient lights the correct path?

(A) 74

(B) 703 r7

(C) 704

(D) 740



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